Check for updates

Received: 17 March 2021 | Revised: 5 June 2021 | Accepted: 8 June 2021

DOI: 10.1111/cen.14549

ORIGINAL ARTICLE

WILEY

Outcomes of 756 patients with differentiated thyroid cancer and excellent response to treatment: An evidence-based paradigm for long-term surveillance strategies

Khyatisha Seejore[1,2] | Omar Mulla[3] | Georgina E. Gerrard[4] | Vanessa M. Gill[4] | Ahmed Al-Qaissi[1] | James W. Moor[5] | Robert D. Murray[1,2]

[1]Department of Endocrinology, Leeds Centre for Diabetes and Endocrinology, St James's University Hospital, Leeds Teaching Hospitals NHS Trust, Leeds, UK

[2]Division of Cardiovascular and Diabetes Research, Leeds Institute of Cardiovascular and Metabolic Medicine (LICAMM), University of Leeds, Leeds, UK

[3]Department of ENT Surgery, Doncaster Royal Infirmary, Doncaster, UK

[4]Department of Clinical Oncology, Leeds Cancer Centre, St James's University Hospital, Leeds Teaching Hospitals NHS Trust, Leeds, UK

[5]Department of ENT Surgery, Leeds General Infirmary, Leeds Teaching Hospitals NHS Trust, Leeds, UK

Correspondence
Robert D. Murray, Department of Endocrinology, Leeds Centre for Diabetes and Endocrinology, St James's University Hospital, Leeds Teaching Hospitals NHS Trust, Beckett St, Leeds LS9 7TF, UK.
Email: robertmurray@nhs.net

Abstract

Background: The 2014 British Thyroid Association thyroid cancer guidelines recommend lifelong follow-up of all thyroid cancer patients. This is probably unnecessary, particularly for differentiated thyroid cancer (DTC) patients with an excellent response to treatment and places significant demand on health service resources.

Design: Single centre retrospective cohort analysis of patients diagnosed and treated at the Leeds Cancer Centre between 2001 and 2014.

Patients: A total of 756 patients were dynamically risk-stratified (DRS) as having 'excellent response to treatment' after total thyroidectomy and radioiodineremnant ablation (RRA) for DTC.

Results: Median follow-up was 11.2 (range: 6.5–18.5) years. Radiological recurrence occurred in 15/756 (2.0%) patients and was always preceded by a raised thyroglobulin or thyroglobulin antibody level. The vast majority of tumour recurrences (13/15, 85%) were identifiable within 5 years of diagnostic surgery. Patients classified as having high-risk disease as per American Thyroid Association (ATA) guidelines had an almost threefold higher recurrence rate (2/34 [5.9%] vs. 13/722 [1.8%]) than those with ATA low-risk or intermediate-risk disease. Tumour histology subtype was a significant contributing factor, with Hürthle cell cancer having a worse prognosis than papillary thyroid cancer (PTC) (5/68 [7.4%] vs. 9/582 [1.5%]; relative risk: 4.76 [95% confidence interval: 1.64–13.8]).

Conclusions: The recurrence rate of DRS patients with excellent response to treatment is low. It is reasonable to consider discharge of ATA low-risk or intermediate-risk patients with PTC who remain disease-free after 5 years of secondary care follow-up. Lifelong follow-up, however, currently remains the standard for subgroups at greater risk.

KEYWORDS
differentiated, follow-up, thyroglobulin, thyroid cancer

 © 2021 John Wiley & Sons Ltd.

1 | INTRODUCTION

Thyroid cancer is the most common endocrine neoplasm worldwide. Differentiated thyroid cancer (DTC) is classified as either papillary thyroid cancer (PTC) or follicular thyroid cancer (FTC), including oncocytic follicular (Hürthle) cell carcinoma. The incidence of DTC is increasing globally,[1] due to an increase in small PTC discovered incidentally during radiological investigations or thyroid surgery for benign conditions.[2] The long-term outcome for patients treated effectively for DTC is usually favourable, with DTC-related deaths occurring primarily in the few patients with more advanced disease.

In the United Kingdom, management of thyroid cancer follows specialist multidisciplinary team decisions in accordance with guidelines from the British Thyroid Association (BTA).[3] Use of prognostic scoring and staging systems, as well as accurate risk stratification of patients, are recommended to tailor treatment and follow-up strategy. Postoperative risk stratification adapted from the American Thyroid Association (ATA) informs the probability of persistent or recurrent disease at a pretreatment stage of the patient pathway. The 2009 ATA initial risk stratification system classifies DTC patients into high-, intermediate- and low-risk cohorts calculated from each patient's disease stage following thyroidectomy.[4] High-risk (HR) ATA patients demonstrate evidence of gross extra-thyroidal extension (T4 disease), presence of distant metastases (M1 disease status) or incomplete macroscopic tumour resection (R2 margin status). In the revised 2015 ATA classification, the presence of lymph node metastasis larger than 3 cm was added to the above criteria.[5]

Dynamic risk stratification (DRS) was introduced in the 2014 BTA guidelines for DTC patients who undergo total thyroidectomy and subsequent radioiodine remnant ablation (RRA) to define the initial response to treatment and guide further management. The 9–12 months postradioiodine DRS is based upon the combined outcome of radiological and biochemical data. At this time point, stimulated thyroglobulin (sTg) less than 1.0 μg/L in the absence of structurally identifiable disease, classifies patients as having an excellent response to treatment with low risk (LR) of recurrence. In these patients, the degree of TSH suppression can be relaxed, aiming for the lower half (0.3–2.0 mU/L) of the reference range.[6,7]

The aim of this study is to identify key prognostic factors important for defining surveillance protocols and assess whether a subset of these patients can be safely discharged from secondary care.

2 | MATERIALS AND METHODS

2.1 | Ethical considerations

This study was designed as a clinical service review to assess current practice and improve patient outcomes. As the study was undertaken using retrospective deidentified data, no informed consent was requested from the patients. The study protocol was in accordance with relevant national guidelines and regulations.

2.2 | Study population

A retrospective analysis of all consecutive DTC patients who received total thyroidectomy and RRA between 2001 and 2014 at the Leeds Cancer Centre (LCC) was undertaken. The LCC is a single tertiary centre with a catchment area of ~3 million. Thyroid cancer follow-up alternates between the LCC and peripheral cancer units. All patients were managed according to BTA guidelines current at the time of treatment.

Inclusion criteria were (1) histologically confirmed diagnosis of PTC, FTC or HTC thyroid cancer treated by total thyroidectomy (single- or two-stage) with adjuvant RRA within 6 months of the definitive surgical procedure; (2) DRS performed within 9–12 months following RRA; (3) excellent response to treatment on DRS and (4) subsequent follow-up with periodic thyroglobulin and thyroglobulin antibody (TgAb) measurements for a minimum of 5 years posttreatment. Exclusion criteria were (1) patients with non-DTC histological subtypes (medullary, anaplastic thyroid cancer) and poorly differentiated DTC (PDTC) and (2) DTC patients lost to follow-up. Tumours were graded according to TNM Edition 7.

2.3 | DRS and retrospective ATA classification

Biochemical DRS was carried out using recombinant human TSH (rTSH). At 9–12 months post-RRA, rTSH 0.9 mg was administered intramuscularly on Days 1 and 2, followed by measurement of sTg and TgAbs on Day 5. Radiological DRS was performed simultaneously. If (1) sTg was less than 1.0 μg/L; (2) TgAbs were undetectable and (3) imaging (neck ultrasound and/or radioiodine [RAI] uptake scan) showed no evidence of residual thyroid tissue, patients were placed in the 'excellent response to treatment' group (defining them as having LR of disease recurrence) and therefore included in this study. DRS as per the BTA 2014 guidelines was assigned many years after RRA in some cases.

After the study cohort was identified, the initial ATA risk estimate valid at the time of thyroidectomy was retrospectively applied to categorise each individual patient. ATA LR patients were defined as having intrathyroidal DTC with no evidence of extra-thyroidal extension, vascular invasion or metastases. Intermediate-risk (IR) patients demonstrated either microscopic extra-thyroidal extension, cervical lymph node metastases less than 3 cm in diameter, RAI-avid disease in the neck outside the thyroid bed, vascular invasion or aggressive tumour histology. HR patients had gross extra-thyroidal extension, incomplete tumour resection, distant metastases, inappropriate postoperative serum thyroglobulin values or metastatic ≥ 3 cm lymph node involvement.[5]

Patients were grouped according to their combined ATA and DRS risk status and are analysed in two distinct cohorts; (1) ATA LR and IR with DRS excellent response to treatment (Group A) and (2) ATA HR with DRS excellent response to treatment (Group B).

2.4 | Disease recurrence

Biochemical recurrence was defined by raised serum Tg greater than 1.0 μg/L or elevated TgAb levels after an undetectable Tg disease-free period. An immunometric conventional assay, the Immulite 1000, was used to measure thyroglobulin with a functional sensitivity of less than 0.9 μg/L. Highly sensitive thyroglobulin assays, with a functional sensitivity of less than 0.1 μg/L, were not available in the Leeds region during the study period (2001–2013). Radiological recurrence was defined as recurrent thyroid carcinoma on imaging. For the purpose of this study, patients with isolated raised Tg or TgAb in the absence of structural disease on follow-up imaging were considered separately.

2.5 | Statistical analysis

Patient characteristics are reported as median and ranges (continuous variables) or absolute numbers and percentages (nonparametric variables). Intergroup differences involving these variables were evaluated with the Fisher exact test. Differences with a $p < .05$ were considered statistically significant. All analyses were performed with SPSS software (version 14.0).

3 | RESULTS

3.1 | Demographics

A total of 756 patients treated with total thyroidectomy followed by RRA had an excellent response to treatment upon DRS as per BTA guidelines (Table 1). Most were histologically PTC (77%), with FTC and HTC accounting for 14% and 9%, respectively. The median duration of follow-up was 11.2 years. Radiological recurrence at any site occurred in 15 (2.0%) patients. Histologically, 60% of recurrences were PTC, 33% HTC and 7% FTC. The relative risk (RR) of recurrence was significantly higher amongst patients with HTC as compared to PTC (5/68 [7.4%] vs. 9/582 [1.5%]; RR: 4.76 [95% confidence interval: 1.64–13.78]).

Applying the postoperative ATA risk stratification system relevant at the time of surgery, 722 (96%) of the patients were found to have ATA LR disease and ATA IR disease and 34 patients (4%) had ATA HR disease. The findings are summarised in Figure 1.

3.2 | ATA LR and IR with DRS excellent response to treatment (Group A)

The majority had histologically PTC (76%), with FTC and HTC accounting for 15% and 9%, respectively. The median duration of follow-up was 10.3 years (Table 1).

Disease recurrence was evident in 1.8% of patients; 54% of whom were PTC, 38% were HTC and 8% FTC (Table 1).

TABLE 1 Characteristics of the overall cohort and subcohorts after stratification by ATA classification

	Overall cohort	ATA classification	
		Low/Intermediate Risk (Group A)	High Risk (Group B)
I. **Baseline**			
n	756	722	34
Female, *n*	544	519	25
Age at diagnosis, years	46 (17–91)	46 (17–91)	45 (18–86)
Histological subtype, *n* (%)			
PTC	562 (77%)	550 (76%)	32 (94%)
FTC	106 (14%)	106 (15%)	0
HTC	68 (9%)	66 (9%)	2 (6%)
Duration of follow-up, years	11.2 (6.5–15.5)	10.3 (5.0–17.5)	10.3 (5.0–17.5)
II. **Recurrences**			
n	15 (2%)	13 (1.8%)	2 (6%)
Age at recurrence, years	64 (45–89)	62 (46–89)	–
Histological subtype, *n* (%)			
PTC	9 (60%)	7 (54%)	2 (100%)
FTC	1 (7%)	1 (8%)	0
HTC	5 (33%)	5 (38%)	0

Abbreviations: ATA, American Thyroid Association; FTC, follicular thyroid carcinoma; HTC, Hürthle cell carcinoma; PTC, papillary thyroid carcinoma.

Biochemical recurrence preceded radiological recurrence in all patients, with a median delay of 7 (range: 1–58) months between these two diagnostic tests proving positive for recurrent disease. The minimum duration to biochemical recurrence was 19 months after DRS, with a median time of 40 (range: 19–108) months. Recurrent disease was identified locally in the thyroid bed in 23% (3/13), regionally in cervical lymph nodes in 23% (3/13) and at distant sites in seven patients (54%, including five patients with lung metastases). One patient (1/13, 8%) with HTC had more than one site of recurrent disease. One patient had a solitary bone metastasis and another had multiple liver metastases. All but four patients were subjected to further interventions; two were not medically fit to receive any treatment and the other two were placed under active surveillance. Three patients (3/13, 23%) subsequently died due to metastatic thyroid cancer with death occurring 9, 30 and 82 months after recurrence had been biochemically diagnosed. Tumour staging and site of recurrence are shown in Table 2.

It is noteworthy that within this group, a further six patients demonstrated a raised Tg during follow-up, but did not develop structural disease recurrence during surveillance. No TgAbs were

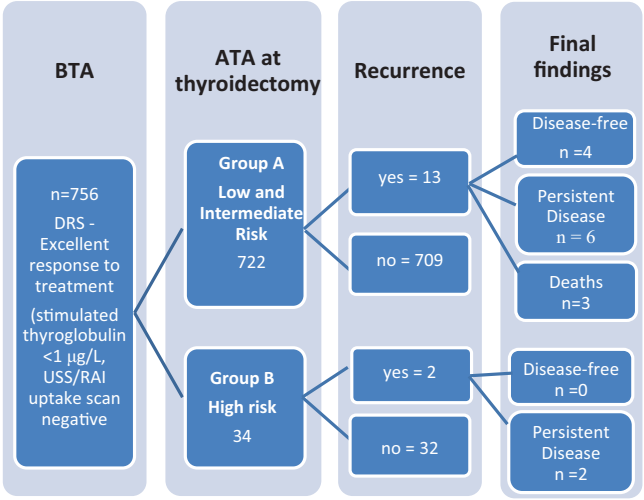

FIGURE 1 Schematic detailing patients allocated to ATA low and intermediate risk and ATA highrisk groups, with diagnoses of recurrence for each group and final findings at the end of follow-up. ATA, American Thyroid Association; BTA, British Thyroid Association; RAI, radioiodine; USS, ultrasound scan [Color figure can be viewed at wileyonlinelibrary.com]

detectable in any of these six patients. In four of these patients, Tg eventually returned to less than 1.0 µg/L without intervention. The remaining two patients still have persistently elevated Tg and have been followed up for 54 and 64 months, respectively, and have no evidence of radiological recurrence to date.

3.3 | ATA HR with DRS excellent response to treatment (Group B)

Thirty-four patients were included in this group (Table 1). Six patients (18%) demonstrated lymph node metastasis greater than 3 cm at diagnosis, and 71% (24/34) of patients had an extra-thyroidal extension at the time of surgery. The vast majority were diagnosed with PTC (94%) and the remaining two patients (6%) HTC.

Six patients (18%) were dynamically risk-stratified as having had an excellent response to treatment after total thyroidectomy and a single episode of RRA. The remaining 28 patients (82%) required additional treatment consisting of further surgery (lateral or central neck dissections, further RAI or external beam radiotherapy) before being categorised as having an excellent response after DRS. Interestingly, four patients diagnosed with M1 disease at presentation (lung and spine metastasis), which required external beam radiotherapy in addition to their total thyroidectomy and RRA, still achieved DRS of excellent response to treatment, hence their inclusion in the study.

Thyroid cancer recurrence in this group occurred in two patients (6%), in both cases biochemical recurrence preceding radiological disease. Both had histologically PTC and disease recurrence occurred outside the thyroid bed. One patient had a total thyroidectomy and selective neck dissection for a pT4a N1b R0 V2 M0 PTC (complete surgical resection but extensive vascular invasion). Biochemical recurrence was detected at 59 months after completion of treatment, which was then confirmed as pulmonary metastasis by computed tomography scanning 18 months later. She was treated with pulmonary external beam radiotherapy and is currently disease-free, although with a rising Tg concentration. The second patient had a pT3 N1b R0 V1 M0 PTC at staging (complete surgical excision with evidence of vascular invasion). Biochemical recurrence was detected at 36 months and immediately confirmed as regional recurrence in cervical lymph nodes, which was managed by salvage neck dissection. This is summarised in Table 2.

Although the incidence of the recurrent disease appeared to be three times higher in the ATA HR group compared to the ATA LR and IR group, there was no statistically significant difference in the proportion of patients with cancer recurrence between the two groups ($p = .127$).

3.4 | An evidence-based paradigm for long-term surveillance

Following the initial 5 years of surveillance, we detected only two recurrences in the Group A cohort, equating to a recurrence rate of 0.28% (2/722). The data, therefore, support ATA LR and ATA IR patients with PTC who achieve DRS 'excellent response to treatment' can be offered discharge from specialist follow-up as long as they have remained disease-free with undetectable Tg and negative TgAb for 5 years after diagnostic surgery.

By contrast, based on the data presented, there are a number of groups of patients with well-differentiated DTC classified as LR on DRS who we suggest currently require longer, even lifelong follow-up. These include:

1. DTC patients with other histological subtypes, namely FTC or HTC.
2. Patients with ATA HR disease.
3. Patients with ATA LR or ATA IR disease who develop raised thyroglobulin or TgAb levels, or evidence of persistent disease on imaging during the initial 5 years of surveillance.

4 | DISCUSSION

We have shown within a large cohort of 756 DTC patients who achieved an excellent response to treatment at DRS (LR disease), only 15 (2.0%) developed radiological disease recurrence after a median of 11.2 years follow-up. We further analysed the outcomes according to postoperative ATA risk stratification and showed that recurrence rates in both subgroups were low although seemingly higher in the ATA HR compared with ATA LR and IR DTC patients (5.9% vs. 1.8%). We further demonstrated that patients with HTC had an unfavourable prognosis, with a higher risk of tumour recurrence than those with PTC despite achieving an excellent response to treatment.

TABLE 2 Details of all patients diagnosed with recurrent disease

Patient	Tumour stage (TNM)	Histological type	Delay to rise in Tg (months)	Delay from Tg rise to the diagnosis of radiological recurrence (months)	Site of recurrence	Further treatment	Outcome
ATA low or intermediate risk and DRS excellent response to treatment (Group A, n = 722)—all were M0							
1	pT3N0	PTC	48	23	Bone	Radiofrequency ablation	Tg stable
2	pT2Nx	HTC	24	30	Lung	Radioiodine	Tg increasing
3	pT3Nx	FTC	21	27	Lung	Surveillance	Tg increasing
4	pT3Nx	HTC	19	1	Regional nodes, thyroid bed and lung	Radioiodine and external radiotherapy	Died
5	pT2Nx	PTC	40	58	Lung	Radioiodine and lung metastectomy	Tg undetectable
6	pT3Nx	HTC	26	58	Lung	Radioiodine	Died
7	pT2Nx	PTC	26	1	Liver	Nil—not medically fit	Died
8	pT3Nx	HTC	51	7	Thyroid bed	Radiotherapy	Tg stable
9	pT3N1b	HTC	53	1	Regional node	Neck dissection then radioiodine	Tg undetectable
10	pT3N1b	PTC	43	53	Regional nodes	Surveillance only	Tg stable
11	pT3N1a	PTC	40	1	Regional nodes	Neck dissection	Tg stable
12	pT3Nx	PTC	108	3	Thyroid bed	Neck dissection	Tg undetectable
13	pT3N0	PTC	97	1	Thyroid bed	Nil—not medically fit	Tg undetectable
ATA high risk and DRS excellent response to treatment (Group B, n = 34)—both were M0							
1	pT4aN1b	PTC	59	18	Lung	Radioiodine and radiotherapy	Tg increasing
2	pT3N1b	PTC	36	2	Regional nodes	Neck dissection	Tg stable

Abbreviations: FTC, follicular thyroid cancer; HTC, Hürthle cell carcinoma; PTC, papillary thyroid cancer; pTNM, pathological tumour nodal and distant metastasis stage; Tg, thyroglobulin.

Within the ATA LR and IR PTC patients, we observed most recurrences within the initial 5 years, with only 2/722 documented recurrences thereafter.

The current study is the first that is reflective of UK practice as defined by the 2007 and 2014 British Thyroid Association Thyroid Cancer Guidelines. By analysing the data set using DRS, ATA classification and histological subtype, these data allow us to hone down on a subset of individuals who would be suitable for discharge to primary care after an initial period of follow-up within secondary/tertiary care for 5 years. There are no studies or consensus statements that we are aware of that have defined these criteria. The study identifies that DRS staging alone is insufficient when used in isolation to allow safe discharge of patients and that ATA staging and histological subtype are equally important when assessing the risk of recurrence and making this decision. The study additionally highlights areas where subject numbers are smaller, less patient-years of data consequently available and therefore warrant further multicentre research, including data on FTC and HTC subtypes.

Our findings confirm a remarkable prognosis after an excellent response to treatment upon DRS comparable to previous studies that have shown a risk of 1%–4% tumour recurrence over 5–10 years.[8–11] Importantly, biochemical recurrence became evident in all patients before detection of the structural disease on imaging, although unfortunately, only four patients had 'curable' recurrence. Additionally, over 85% of disease recurrences were initially identified within 5 years of follow-up, in keeping with a previous multicenter series.[12] All patients with DRS 'excellent response to treatment' DTC currently undergo lifelong follow-up, placing a considerable burden on healthcare services as well as increasing and prolonging anxiety in the patients themselves. The cost of detection of a recurrence is 6–7 times greater in patients with LR DTC compared with HR disease, due to the lower prevalence of recurrence.[13] Current practice guidelines offer no clear recommendation on implementing an individualised approach to cost-effectiveness surveillance in patients with DTC. In the first instance, it is pertinent to identify markers that will distinguish the small subset with HR tumours requiring close and more prolonged follow-up from those with negligible risks for recurrence.

Upon DRS, an excellent response to treatment is usually achieved in 86%–91% of ATA LR patients, 57%–63% of ATA IR patients and

14%–16% of ATA HR patients.[8,9,14] All patients in the current study had DRS 'excellent response to treatment' DTC. Interestingly, we observed a threefold higher recurrence rate in ATA HR patients compared with ATA LR or IR (5.9% vs. 1.8%), suggesting that until more data are available ATA HR patients should continue long-term active monitoring in a specialist setting. Two further studies[8,15] showed 14% of ATA HR patients who achieved an excellent response to treatment upon DRS experienced disease recurrence after 6 years. The higher recurrence rate compared with our data could be linked to the larger proportion of patients harbouring FTC.[15] The presence of distant metastases at presentation was also observed to increase the risk of persistent/recurrent disease. Only 4/756 (0.5%) of our patients initially presented with distant metastatic disease; none of whom relapsed during follow-up. Nonetheless, this is likely a particularly vulnerable subgroup as these patients underwent additional intensive therapy before being reclassified as DRS 'excellent response to treatment'. Conversely, the risk of recurrence in ATA LR or IR DTC patients was reassuringly low at 1%–2%.

The risks of recurrence also depend in part on the tumour histology and staging. Of particular note from our series is that one-third (5/15) of the recurrences were HTC and in our region only 9% of DTC patients have HTC, suggesting that these individuals are at increased risk of recurrence despite what appears to be excellent initial disease control following treatment. Indeed, HTC has traditionally been regarded as a more aggressive histological subtype of thyroid carcinoma compared with PTC, with more frequent distant metastases and lymph node involvement.[16,17] Similarly, FTC, particularly the widely invasive forms, have poorer prognosis and evidence on follow-up of patients with FTC is limited.[5,18] Although no significant difference in recurrence rates between PTC and FTC was reported amongst ATA HR patients in a recent 2019 series,[15] widely invasive FTCs have been associated with a recurrence risk of 30%–55% and disease-specific mortality as high as 30%.[5] Due to a low number of recurrences in our study and considering the natural history of FTCs, it would be prudent that FTC DTC remains under specialist follow-up until more comprehensive data become available. It is important to note that patients with more aggressive poorly differentiated DTC were excluded from our study.[19] Our findings thus do not apply to patients with PDTC who require active lifelong follow-up. Although our numbers of recurrences are reassuringly small, there are other observations that can be made. From the TNM staging of our patients in whom it recurred, all were T2, T3 or T4, providing further data to suggest that those with T1 disease have an excellent outcome.[20]

The hallmark of an excellent response to treatment upon DRS is a stimulated Tg less than 1.0 ug/L and negative TgAb after RRA.[5] Thereafter, any rise in Tg or TgAb level warrants urgent localisation studies. In the current study, all cases of recurrence were biochemically detected before radiological evidence of disease was identified. A rise in Tg/TgAb levels at any point during follow-up is strongly predictive of tumour recurrence and these patients should be carefully monitored for an indefinite time period. It is noteworthy, however, that we also describe a small group of DTC patients ($n = 4$) with transitory isolated thyroglobulinaemia, which resolved spontaneously without structural evidence of tumour recurrence. Similarly, another two patients who have persistently isolated raised Tg/TgAb for many years without the identifiable disease to date may also turn out having false-positive results, which is acceptable given the high positive predictive value of serum Tg/TgAb.

One of the strengths of this study is the large number of patients with ATA HR DTC reclassified as an excellent response to treatment upon DRS compared with previous studies.[10,12] It is the second-largest series behind that of Van Velsen et al.[15] who also advocate a more cautious approach for this subset of patients. Secondly, our follow-up of 11.2 years provides a reliable timeframe for the detection of tumour recurrence. A possible limitation of the study relates to its retrospective nature. Postoperative ATA risk estimate was applied after review of historical information recorded in case notes. Similarly, DRS as per the BTA 2014 guidelines was assigned many years after RRA in some cases. A prospective study is however unlikely to be performed given a large number of patients and the long duration of follow-up necessary. Furthermore, the data after a prolonged prospective study may be outdated as a consequence of concurrent advances relating to improvements in thyroglobulin assays and the increasing role of genetic markers for diagnosis and individualising treatment and follow-up. A further limitation is the inability to perform multivariate analysis for recurrence and lack of significance for statistical analyses due to a low number of events. The study was however conducted in a single tertiary centre, all patients subject to standardised clinical practice as per a pre-established treatment protocol, with no variation in follow-up schema.

Based on our data we propose that ATA LR or ATA IR patients with PTC who achieve DRS 'excellent response to treatment' can be offered discharge from specialist follow-up as long as they have remained disease-free with undetectable Tg and negative TgAb for 5 years after diagnostic surgery. Combining our data with that in the literature, there are a number of groups of patients with DTC who will require longer, even lifelong follow-up. This includes:

1. DTC patients with other histological subtypes, namely FTC, HTC or PDTC.
2. Patients with ATA HR disease.
3. Those with ATA LR or ATA IR disease who develop raised thyroglobulin or TgAb levels or evidence of persistent disease on imaging.[3]
4. Patients who have received external beam radiotherapy to the neck to monitor for late adverse effects of the irradiation itself.
5. Those with concomitant hypoparathyroidism who require long-term endocrinology input and monitoring.

Importantly, before the discharge of any patients with a diagnosis of DTC, it is important the patient is provided with written and verbal information as to 'red flag' symptoms (i.e. painless neck masses), and that there is an established pathway through which patients can directly be reviewed by the specialist team without the need for primary care rereferral.

In conclusion, this large retrospective cohort analysis of patients with DTC who achieved an excellent response to treatment upon DRS demonstrates excellent clinical outcomes. Recurrent disease is

rare and usually occurs within 5 years of diagnostic surgery. We further highlight the importance of taking into account ATA classification, histological subtype and staging when individualising patient surveillance. Recurrence in those with PTC ATA LR or IR disease is rare after the initial 5 years of follow-up, potentially allowing discharge of these individuals to primary care at that time point.

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

AUTHOR CONTRIBUTIONS

GEG proposed and designed the study and collated the dataset. The original manuscript was written by KS, OM and GEG. All authors were involved throughout the study with data interpretation, review of the data analysis, review and revision of the initial manuscript. RDM provided oversight for the study, collated responses from co-authors and revised the final manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Khyatisha Seejore http://orcid.org/0000-0003-1633-4155
Robert D. Murray http://orcid.org/0000-0001-6730-6109

REFERENCES

1. Lim H, Devesa SS, Sosa JA, Check D, Kitahara CM. Trends in thyroid cancer incidence and mortality in the United States, 1974-2013. *JAMA*. 2017;317(13):1338-1348.
2. Sipos JA, Mazzaferri EL. Thyroid cancer epidemiology and prognostic variables. *Clin Oncol*. 2010;22(6):395-404.
3. Perros P, Boelaert K, Colley S, et al. Guidelines for the management of thyroid cancer. *Clin Endocrinol*. 2014;81(1):1-122.
4. Cooper DS, Doherty GM, Haugen BR, et al. Revised American Thyroid Association management guidelines for patients with thyroid nodules and differentiated thyroid cancer. American Thyroid Association (ATA) Guidelines Taskforce on thyroid nodules differentiated thyroid cancer. *Thyroid*. 2009;19(11):1167-1214.
5. Haugen BR, Alexander EK, Bible KC, et al. American Thyroid Association management guidelines for adult patients with thyroid nodules and differentiated thyroid cancer; the American Thyroid Association Guidelines Task Force on thyroid nodules and differentiated thyroid cancer. *Thyroid*. 2016;26(1):1-133.
6. Reed N, Mallick U. Special issue on thyroid cancer. *Clin Oncol*. 2017;29(5):276-277.
7. Freudenthal B, Williams GR. Thyroid stimulating hormone suppression in the long-term follow-up of differentiated thyroid cancer. *Clin Oncol*. 2017;29(5):325-328.
8. Tuttle RM, Tala H, Shah J, et al. Estimating risk of recurrence in differentiated thyroid cancer after total thyroidectomy and radioactive iodine remnant ablation: using response to therapy variables to modify the initial risk estimates predicted by the new American Thyroid Association staging system. *Thyroid*. 2010;20:1341-1349.
9. Vaisman F, Momesso D, Bulzico DA, et al. Spontaneous remission in thyroid cancer patients after biochemical incomplete response to initial therapy. *Clin Endocrinol*. 2012;77:132-138.
10. Jeon MJ, Kim WG, Park WR. Modified dynamic risk stratification for predicting recurrence using the response to initial therapy in patients with differentiated thyroid carcinoma. *Eur J Endocrinol*. 2018;170(1):23-30.
11. Seejore K, Gerrard GE, Gill V, Murray RD. Can we discharge dynamically risk-stratified low-risk (excellent response to treatment) thyroid cancer patients after 5 years of follow-up? *Clin Oncol*. 2019;31:219-224.
12. Durante C, Montesano T, Torlontano M, et al. Papillary thyroid cancer: time course of recurrences during post surgery surveillance. *J Clin Endocrinol Metab*. 2013;98(2):636-642.
13. Wang LY, Roman BR, Migliacci JC, et al. Cost-effectiveness analysis of papillary thyroid cancer surveillance. *Cancer*. 2015;121(23):4132-4140.
14. Castagna MG, Maino F, Cipri C, et al. Delayed risk stratification, to include the response to initial treatment (surgery and radioiodine ablation), has better outcome predictivity in differentiated thyroid cancer patients. *Eur J Endocrinol*. 2011;165:441-446.
15. Van Velsen EFS, Stegenga MT, van Kemenade FJ, et al. Evaluating the 2015 American Thyroid Association Risk Stratification System in high-risk papillary and follicular thyroid cancer patients. *Thyroid*. 2019;29(8):1073-1079.
16. World Health Organization. *WHO Classification of Tumours of Endocrine Organs. In: Lloyd RV, Osamura RY, Klöppel G, Rosai J, eds. WHO Classification of Tumours*. Vol 10. 4th ed. International Agency for Research on Cancer Publications; 2017.
17. McDonald MP, Sanders LE, Silverman ML, Chan HS, Buyske J. Hürthle cell carcinoma of the thyroid gland: prognostic factors and results of surgical treatment. *Surgery*. 1996;120(6):1000-1004.
18. Grani G, Lamartina L, Durante C, Filetti S, Cooper DS. Follicular thyroid cancer and Hurthle cell carcinoma: challenges in diagnosis, treatment and clinical management. *Lancet Diabetes Endocrinol*. 2017;6:500-514.
19. Ibrahimpasic T, Ghossein R, Carlson DL, et al. Outcomes in patients with poorly differentiated thyroid carcinoma. *J Clin Endocrinol Metab*. 2014;99(4):1245-1252.
20. Sapuppo G, Tavarelli M, Belfiore A, Vigneri R, Pellegriti G. Time to separate persistent from recurrent differentiated thyroid cancer: different conditions with different outcomes. *J Clin Endocrinol Metab*. 2019;104(2):258-265.

How to cite this article: Seejore K, Mulla O, Gerrard GE, et al. Outcomes of 756 patients with differentiated thyroid cancer and excellent response to treatment: An evidence-based paradigm for long-term surveillance strategies. *Clin Endocrinol*. 2021;1-7. https://doi.org/10.1111/cen.14549